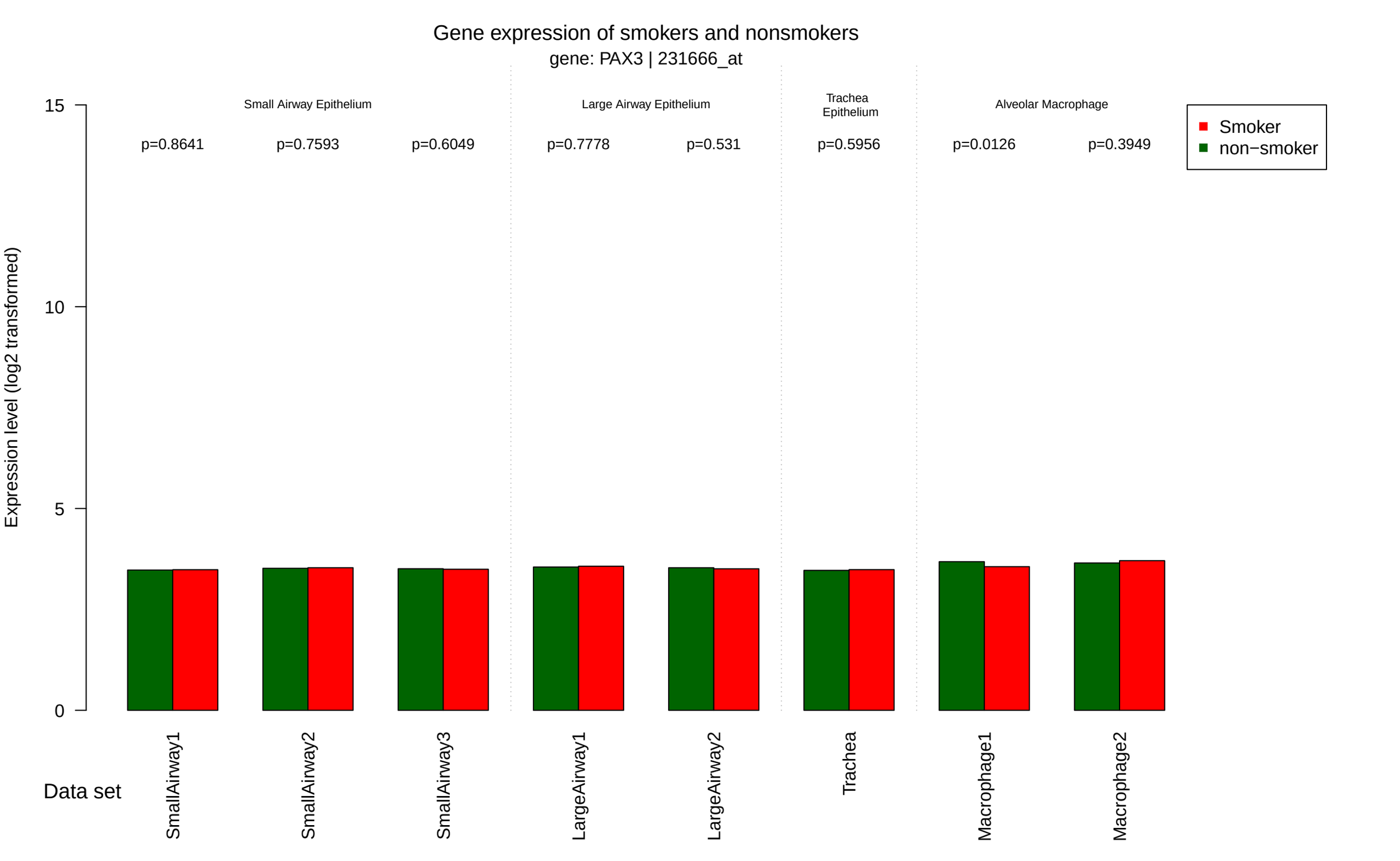
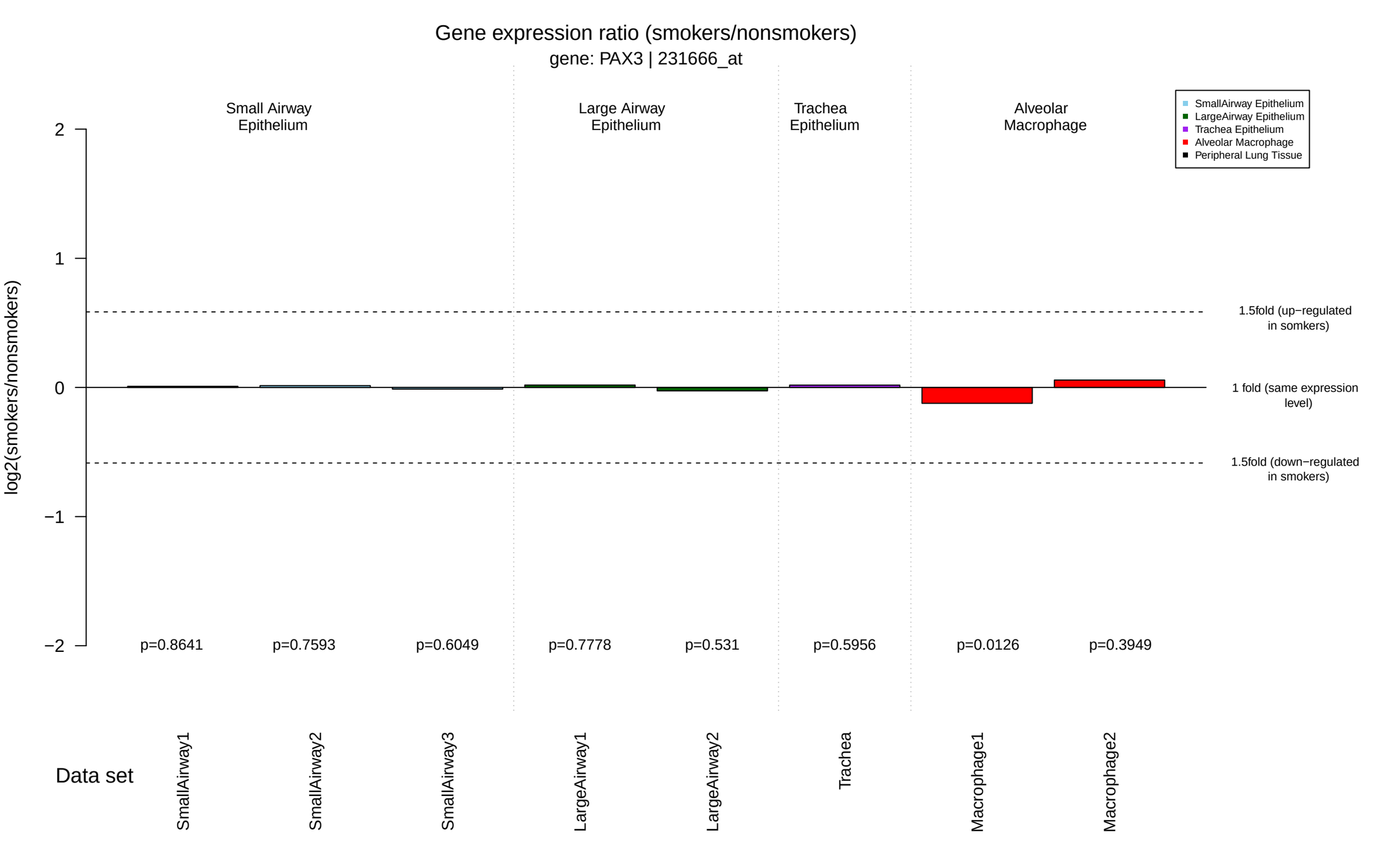
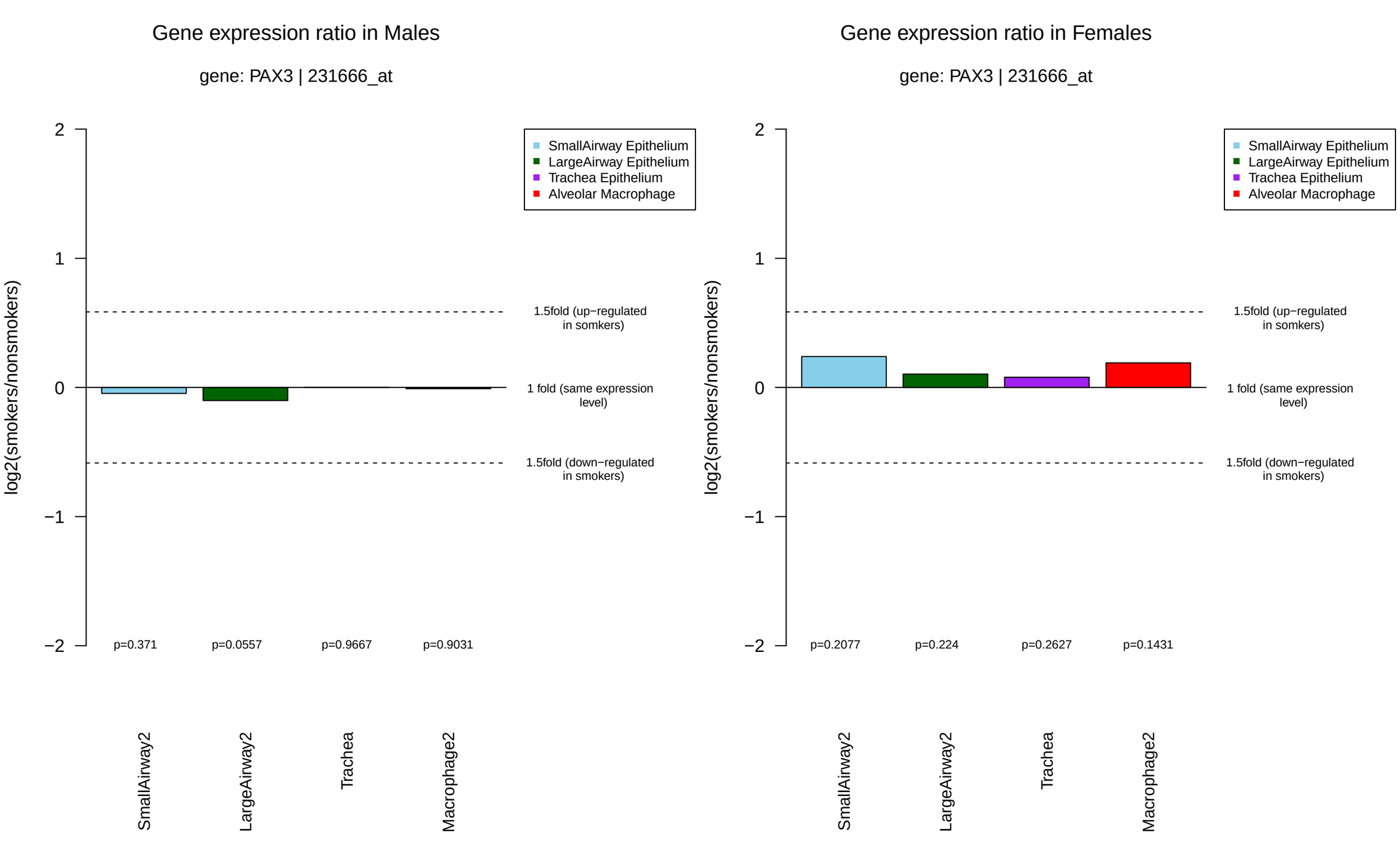
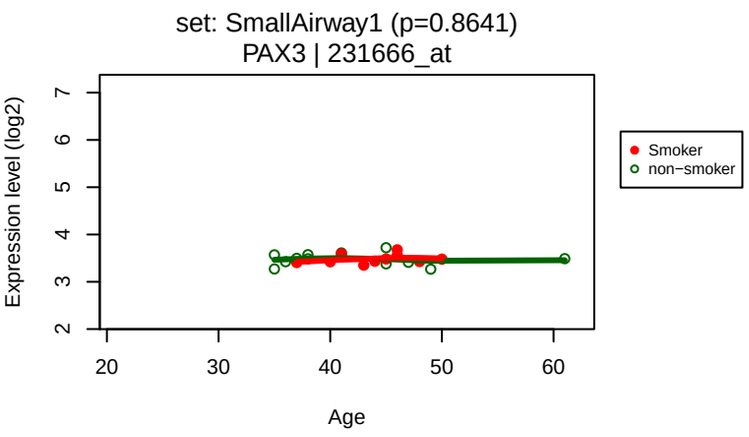
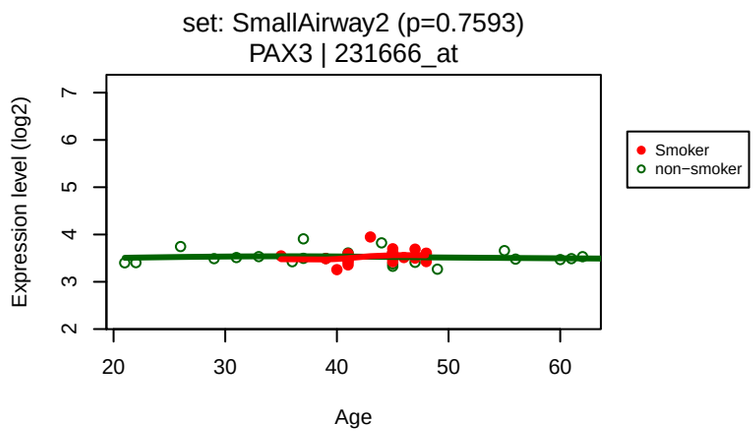
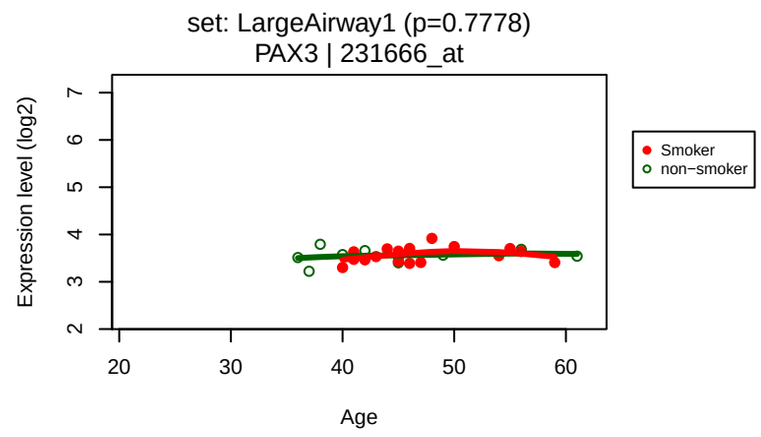

Gene expression of smokers and nonsmokers
gene: PAX3 | 231666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8641
p=0.7593
p=0.6049
p=0.7778
p=0.531
p=0.5956
p=0.0126
p=0.3949
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

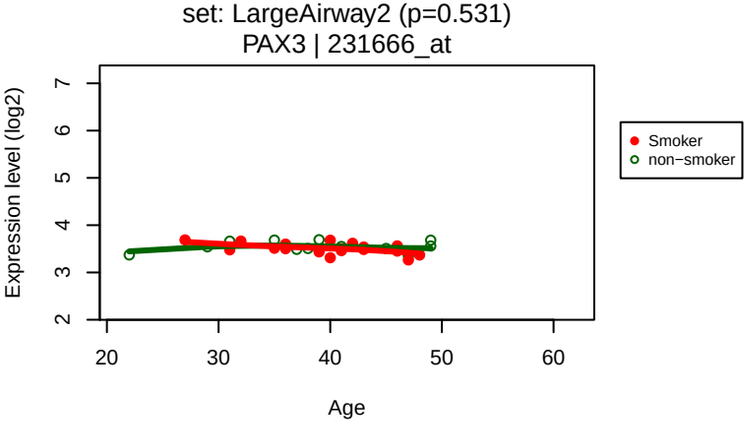

Gene expression ratio (smokers/nonsmokers)
gene: PAX3 | 231666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8641
p=0.7593
p=0.6049
p=0.7778
p=0.531
p=0.5956
p=0.0126
p=0.3949
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

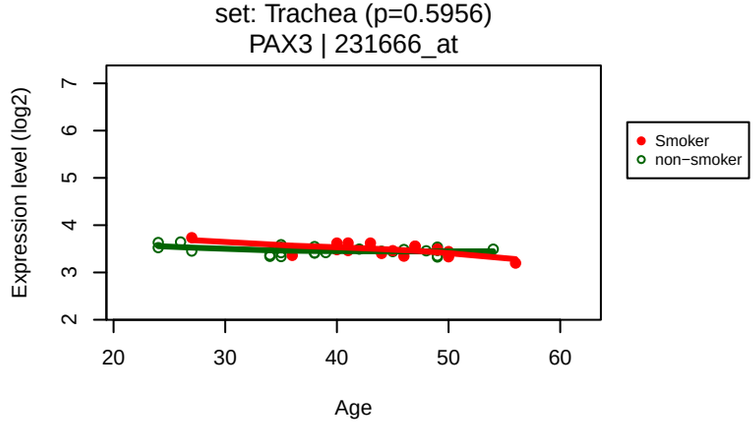

Gene expression ratio in Males
gene: PAX3 | 231666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.371
p=0.0557
p=0.9667
p=0.9031
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PAX3 | 231666_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2077
p=0.224
p=0.2627
p=0.1431
SmallAirway2
LargeAirway2
Trachea
Macrophage2

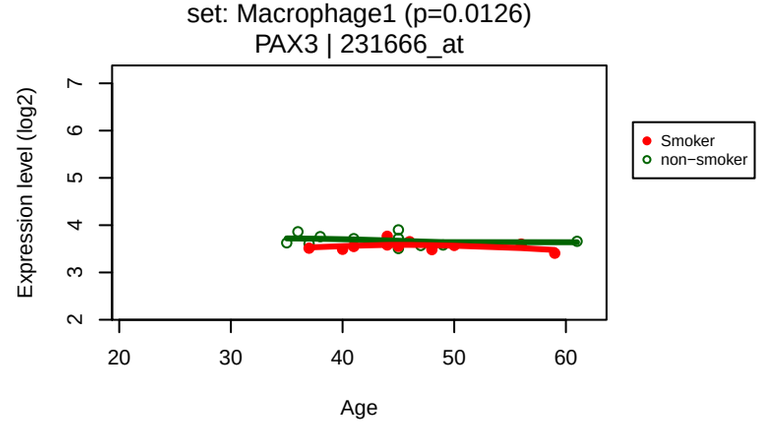

set: SmallAirway1 (p=0.8641)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker

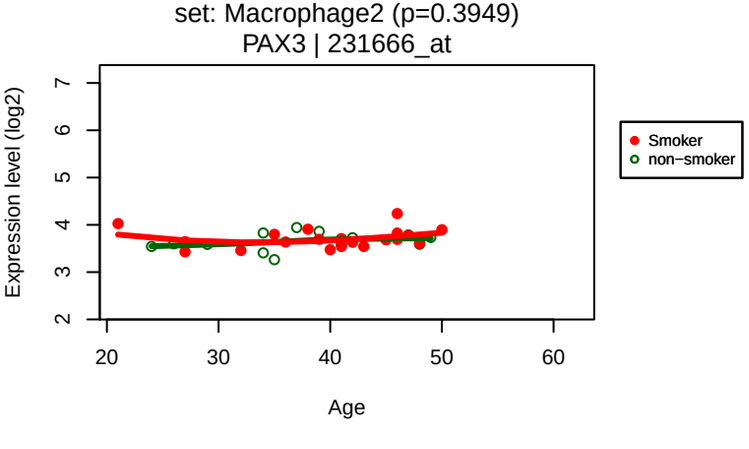

set: SmallAirway2 (p=0.7593)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker

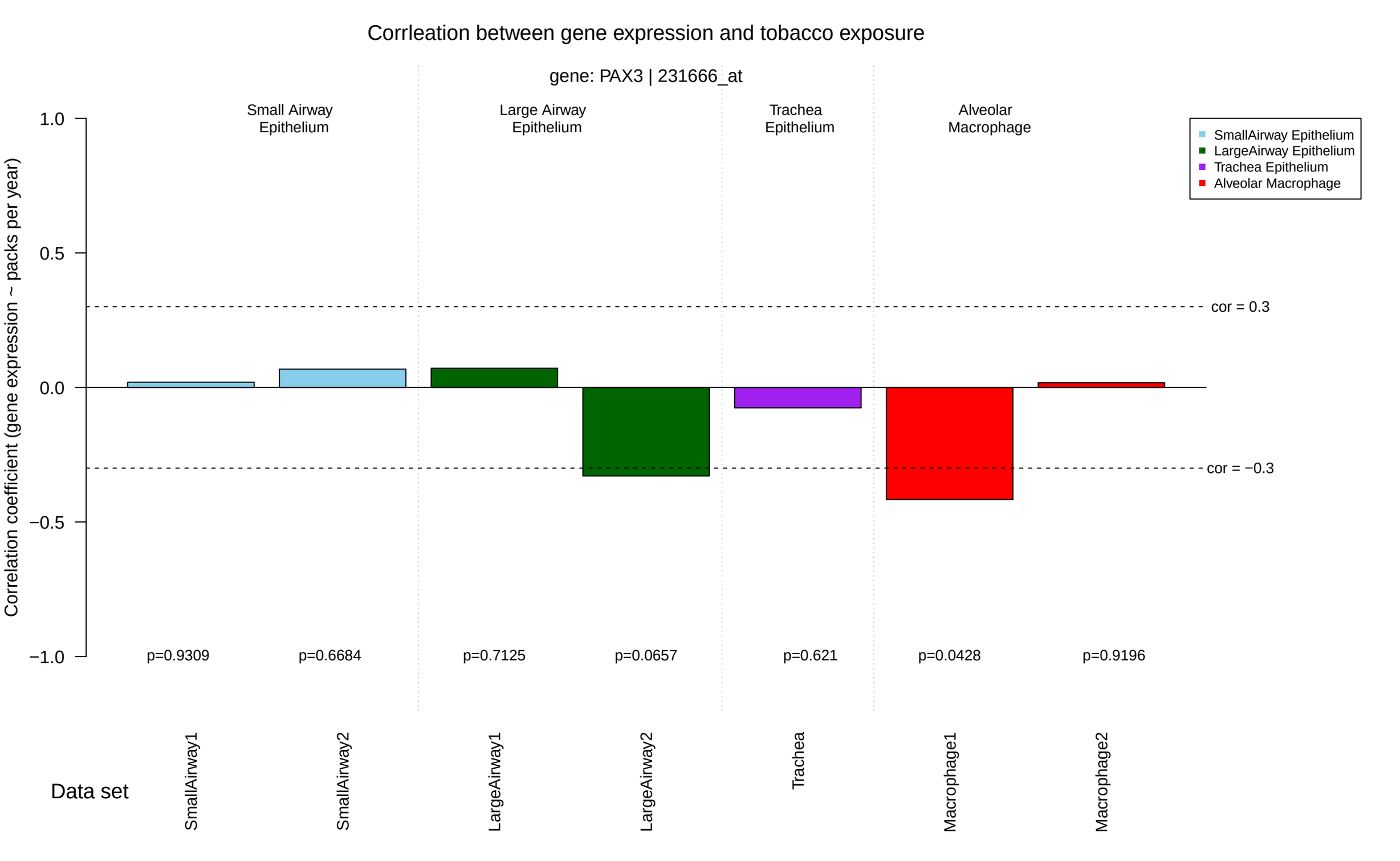

set: LargeAirway1 (p=0.7778)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.531)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5956)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0126)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3949)
PAX3 | 231666_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PAX3 | 231666_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9309
p=0.6684
p=0.7125
p=0.0657
p=0.621
p=0.0428
p=0.9196
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2